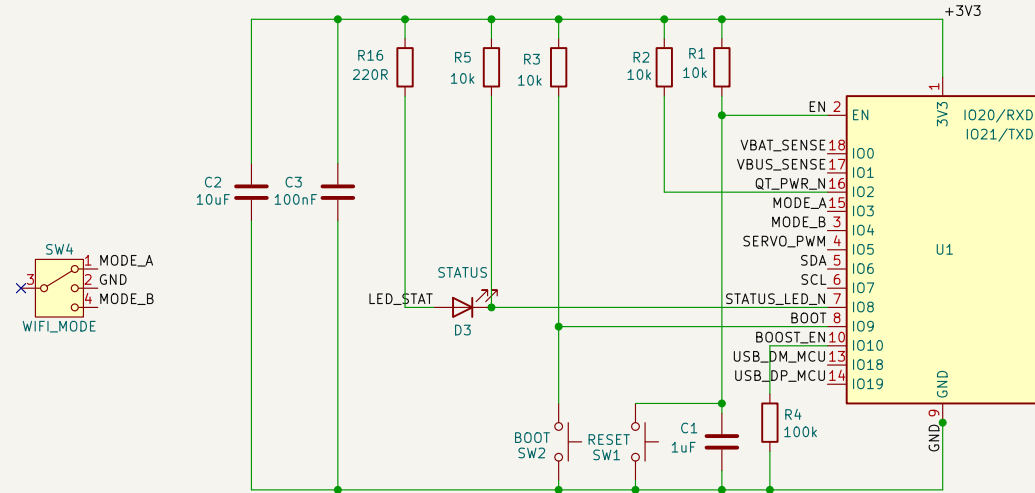


MCU - ESP32-C3-WROOM-02



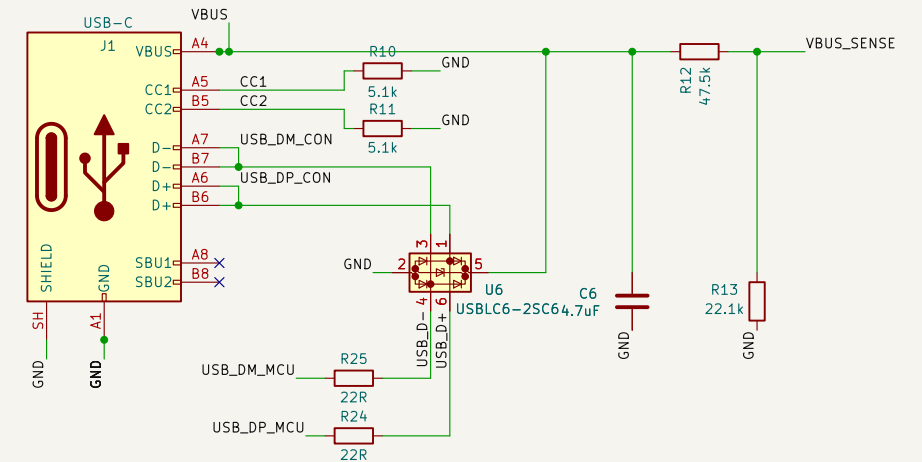
STEMMA QT port (v7): J5 1 GND, 2 3V3, 3 SDA, 4 SCL. Q8 switches it from IO2 (low = on; R2 keeps it off at reset/sleep), Q9/Q10 isolate the bus when off, R29/R30 port pull-ups, U8 ESD, BME688 now at 0x76 (SDO = GND) - 0x77 free for add-ons, R31/R32: charger STAT (5 V) -> IO20 CHG_DET. Low = charging (with VBUS), high = done.

ESP32-C3 notes (MCU box):
BOOT (SW2) low at reset -> SPI boot.
IO8 drives the STATUS LED active-low (LED off = strap high).
IO10 = BOOST_EN; R4 100k keeps the boost off at reset/sleep.
EN: R1 10k + C1 1uF RC delay (Espressif HW design guide).
USB: IO18/IO19 native USB Serial/JTAG - no UART bridge.

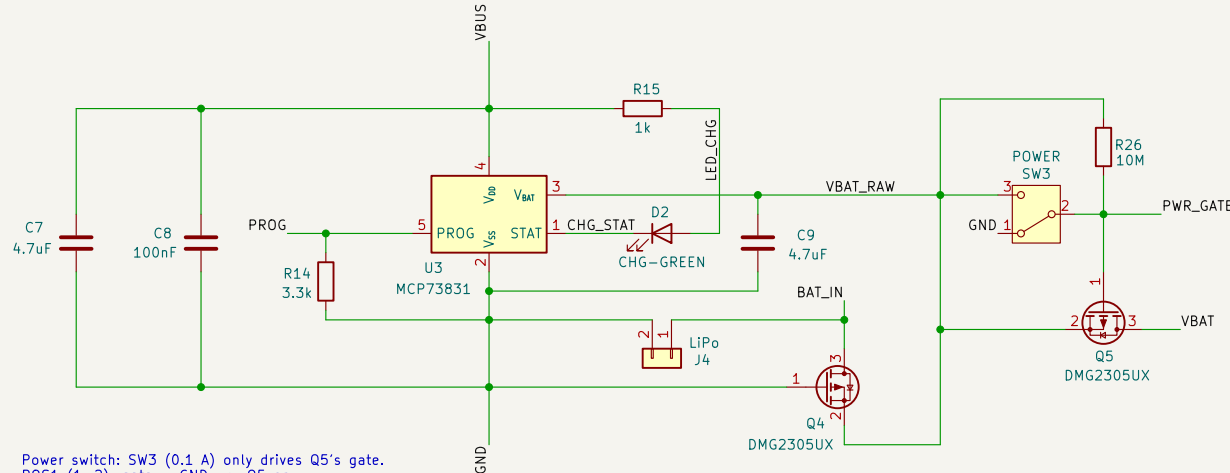
SW4 (OS103011MA7QP1, common = pin 2):
POS1 MODE_A (IO3) low = OFF | POS2 both high = SPARSE | POS3 MODE_B (IO4) low = CONTINUOUS.

Internal pull-ups only. IO3/IO4 are deep-sleep wake-capable (GPIO0-5). VBAT_SENSE IO0 / VBUS_SENSE IO1 = ADC1.

USB-C Input + USB data + ESD + VBUS sense

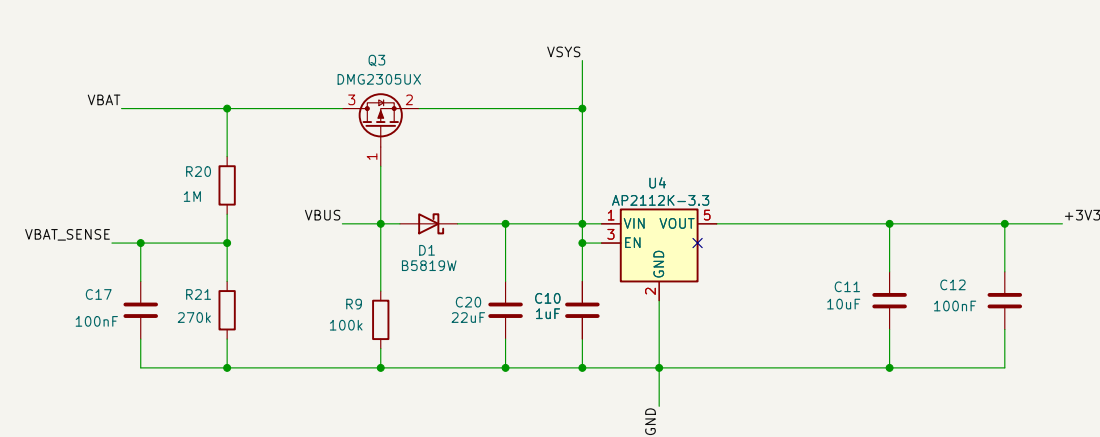


LiPo Charger (MCP73831) + power switch

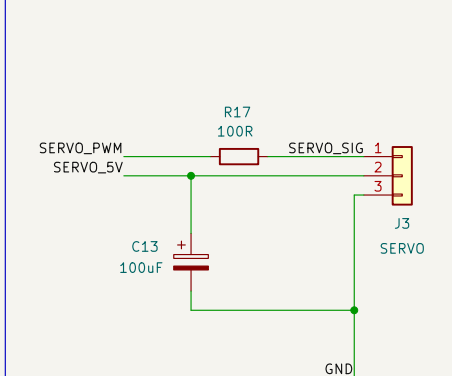


Power switch: SW3 (0.1 A) only drives Q5's gate.
POS1 (1-2): gate = GND -> Q5 on
POS2 (2-3): gate = source -> Q5 off
R26 10M holds Q5 off in the break-before-make gap.

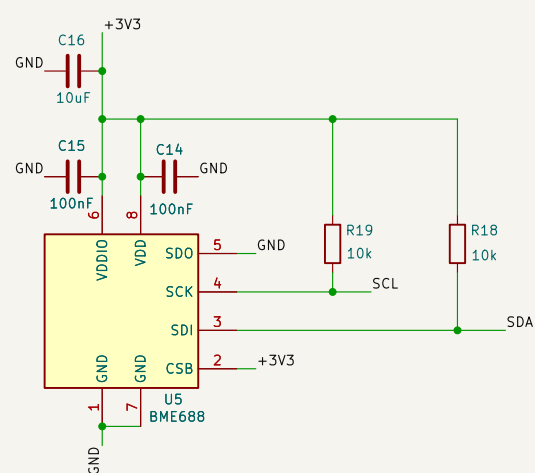
Power Path + 3V3 LDO



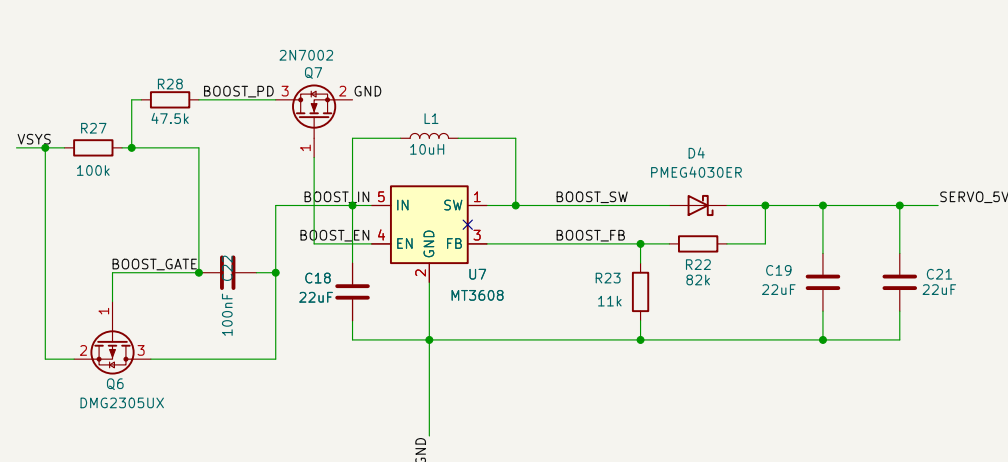
Servo Output



Air Sensor (BME688)



Servo 5V Boost (MT3608)



Layout: hot loop U7 SW -> D4 -> C19/C21 -> GND -> U7 pin 2 within ~8x8 mm on F.Cu. C18 (22 uF) at U7 pin 5, L1 at pin 1, R22/R23 at pin 3. No signals under L1 / the SW node.

Servo rail load switch (v7). A boost converter is not off when EN is low: VSYS -> L1 -> D4 still feeds SERVO_5V (servo idle + R22/R23). Q6 disconnects BOOST_IN from VSYS while BOOST_EN (IO10) is low. Q7 pulls the gate to -VSYS-2.5 V via R28 when BOOST_EN is high. C22 (gate-drain) slews BOOST_IN at ~0.5 V/ms -> ~80 mA inrush into ~166 uF downstream. Firmware: BOOST_EN high >= 100 ms before a move.

Power Flags



ESP32-C3 + BME688 + SG92R (v7)

Sheet: /
File: air-quality-pcb-v7.kicad_sch

Title: Air Quality Checker

Size: A3
KiCad E.D.A. 10.0.6

Date:
Rev: G
Id: 1/1